

EXPLAINABLE BANK ACCOUNT FRAUD DETECTION USING MACHINE LEARNING

CAPSTONE PROJECT

Postgraduate Diploma in Artificial Intelligence and Machine Learning
Asian Institute of Management, School of Executive Education and Lifelong Learning

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ABSTRACT

This capstone develops an explainable machine-learning workflow for prioritizing potentially fraudulent bank-account applications. The Bank Account Fraud Dataset Suite - NeurIPS 2022 was used with Base.csv as the primary development dataset and Variant II.csv as an external robustness stress test. Both datasets contain 1,000,000 records, 32 original variables, and a fraud prevalence of 1.103%, making the problem a highly imbalanced binary-classification task.

The analysis followed a temporal design: months 0-5 were used for training, month 6 for validation and threshold selection, and month 7 for final testing. Logistic Regression, Decision Tree, Random Forest, and XGBoost were tuned using PR-AUC/Average Precision. XGBoost achieved the strongest cross-validated PR-AUC (0.140452), but Logistic Regression generalized better to the independent validation month and was selected as the provisional champion with validation PR-AUC 0.164799. On the Base month 7 test set, it achieved PR-AUC 0.198539, ROC-AUC 0.890668, and captured 67.02% of fraud within the highest-risk 10% of applications. The same model achieved PR-AUC 0.201282 and top 10% recall 66.46% on Variant II without retraining.

Explainability and responsible-AI analysis identified material differences across age, income, employment, and housing groups, as well as probability-calibration and threshold-transfer concerns. The findings support use of the model as a human-supervised review-prioritization tool, not as an automated rejection mechanism. The project concludes with mitigation experiments, governance recommendations, reproducible model artifacts, and a prioritized improvement roadmap.

EXECUTIVE SUMMARY

This capstone project developed an explainable machine-learning approach to support fraud detection in bank account applications. Using the Bank Account Fraud Dataset Suite, the project treated fraud detection as a supervised binary-classification problem and compared multiple models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost. Although XGBoost achieved the highest cross-validated PR-AUC of 0.140452, Logistic Regression was selected as the provisional champion because it performed best on the independent validation set while remaining more interpretable and easier to govern. On the Base test set, the model achieved a PR-AUC of 0.198539, ROC-AUC of 0.890668, and captured 67.02% of fraudulent applications within the top 10% highest-risk applications, demonstrating strong value as a fraud-review prioritization tool.

The results show that the model can materially improve manual review efficiency by concentrating a large share of fraud cases into a smaller review population, with an illustrative estimated net benefit of 6,419,500.00. The Variant II stress test also showed stable ranking performance, with a PR-AUC of 0.201282 and 66.46% recall at the top 10%, although threshold behavior and calibration varied across datasets. Therefore, the recommended use of the model is as a human-supervised decision-support tool, not as an automatic approval or rejection system. Before any real-world deployment, the model should be validated using actual banking data, recalibrated for local fraud prevalence and review capacity, and continuously monitored for fairness, calibration, data drift, and operational performance.

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1. PROBLEM UNDERSTANDING AND FRAMING

Fraudulent bank account applications create financial, operational, regulatory, and reputational risks for financial institutions. Once opened, fraudulent accounts may be used for identity fraud, mule activity, unauthorized transactions, money laundering, and other illicit financial activities. These risks are especially relevant in digital account opening, where applications can be submitted quickly and fraud patterns may evolve faster than traditional rule-based controls can detect.

The business problem is that fraud teams have limited review capacity and cannot manually investigate every application in depth. The data science problem is therefore to estimate whether an account application is fraudulent or legitimate using available application, behavioral, device, session, credit, and time-related features. The model output is a fraud-risk score that ranks applications from highest to lowest risk.

This capstone is framed as a supervised binary-classification task. The target variable is `fraud_bool`, where 1 represents a fraudulent bank-account application and 0 represents a legitimate application. The model is explicitly positioned as a human-supervised decision-support tool. It is not designed to make automatic approval or rejection decisions.

Metric / KPI	Purpose in the Capstone
PR-AUC / Average Precision	Primary metric because fraud is rare and minority-class ranking matters.
ROC-AUC	Supporting measure of overall discrimination.
Recall and Recall at Top 10%	Measures fraud captured under limited investigation capacity.
Precision and False Positive Rate	Measures analyst workload and potential customer friction.
Estimated Net Benefit	Translates fraud capture, review cost, and false-positive cost into business value.
Fairness and Calibration Metrics	Assess responsible AI, subgroup behavior, and risk of unreliable scores.

Table 1. Success Metrics and Business KPIs

2. DATA COLLECTION AND UNDERSTANDING

The project used the Bank Account Fraud Dataset Suite - NeurIPS 2022. Base.csv was used as the primary modeling dataset because it represents the baseline BAF variant. Variant II.csv was used as a separate robustness and fairness stress-test dataset. The two datasets were intentionally not merged because each variant represents a different controlled data setting.

Dataset	Rows	Columns	Fraud Count	Fraud Rate	Missing Cells	Duplicate Rows	Months
Base.csv	1,000,000	32	11,029	1.103%	0	0	0-7
Variant II.csv	1,000,000	32	11,029	1.103%	0	0	0-7

Table 2. Dataset Overview

The datasets have the same column structure, no missing cells, no duplicate rows, 27 numeric columns, and 5 categorical columns. The categorical variables include payment type, employment status, housing status, application source, and device operating system. Although the data is clean from a missing-value and duplicate perspective, the extremely low fraud prevalence makes the modeling problem difficult and requires metrics that focus on the fraud class rather than overall accuracy.

Data Segment	Rows	Fraud Count	Fraud Rate
Base training, months 0-5	794,989	8,151	1.0253%
Base validation, month 6	108,168	1,450	1.3405%
Base final test, month 7	96,843	1,428	1.4746%
Variant II robustness test, month 7	96,842	1,428	1.4746%

Table 3. Temporal Split Used for Model Development and Testing

The temporal split was important because the model is intended to learn from past applications and score later applications. Using month 6 for validation and month 7 for final testing provides a more realistic assessment than a random split alone.

3. DATA PREPROCESSING, APPLIED EDA, AND FEATURE ENGINEERING

The preprocessing strategy was designed to prepare raw datasets for modeling while preserving fraud-relevant behavioral signals. The workflow used median imputation for numeric fields, most-frequent imputation for categorical fields, robust scaling, and one-hot encoding with unknown-category handling. All transformations were implemented within modeling pipelines to reduce leakage and ensure the same logic was applied consistently during validation and testing. Feature engineering expanded the Base dataset from 32 original variables to 53 columns. The engineered features captured address-history indicators, balance-behavior flags, device and email patterns, velocity ratios, proposed-credit-limit transformations, and age or credit-risk bands. Variant II was processed using the same feature logic and retained as a separate robustness dataset.

Applied EDA confirmed severe class imbalance, temporal variation in fraud prevalence, and differences in behavioral and categorical distributions. Mutual-information analysis, Random Forest feature importance, and PCA were used as analytical lenses for feature selection and dimensionality understanding. PCA was retained for diagnostic analysis rather than final model deployment because interpretability was more important than dimensional compression for this fraud-risk use case.

Preprocessing or EDA Component	Purpose
Missing and duplicate checks	Confirmed that both datasets had no missing cells or duplicate rows.
Temporal train-validation-test split	Reduced the risk of overly optimistic performance from randomly mixed time periods.
Robust scaling and one-hot encoding	Prepared numeric and categorical variables for linear and tree-based models.
Domain-derived feature engineering	Captured fraud-relevant device, velocity, address, credit, and session patterns.
Feature importance and PCA	Supported feature understanding, dimensionality assessment, and explainability.

Table 4. Preprocessing, EDA, and Feature Engineering Summary

4. MODEL IMPLEMENTATION AND COMPARISON

Step 4 trained and compared Logistic Regression, Decision Tree, Random Forest, and XGBoost using a 150,000-record development sample and stratified three-fold cross-validation. PR-AUC was used as the primary tuning metric because fraud represented only 1.0253% of the training data.

Model	Search Method	Best CV PR-AUC	Key Notes
Logistic Regression	GridSearchCV	0.137709	Converged with newton-cg; selected C=0.02.
Decision Tree	RandomizedSearchCV	0.057354	Lowest PR-AUC and weaker minority-class ranking.
Random Forest	RandomizedSearchCV	0.122551	Nonlinear model but slower and weaker than Logistic Regression and XGBoost.
XGBoost	RandomizedSearchCV	0.140452	Best cross-validated PR-AUC.

Table 5. Cross-Validated Model Comparison

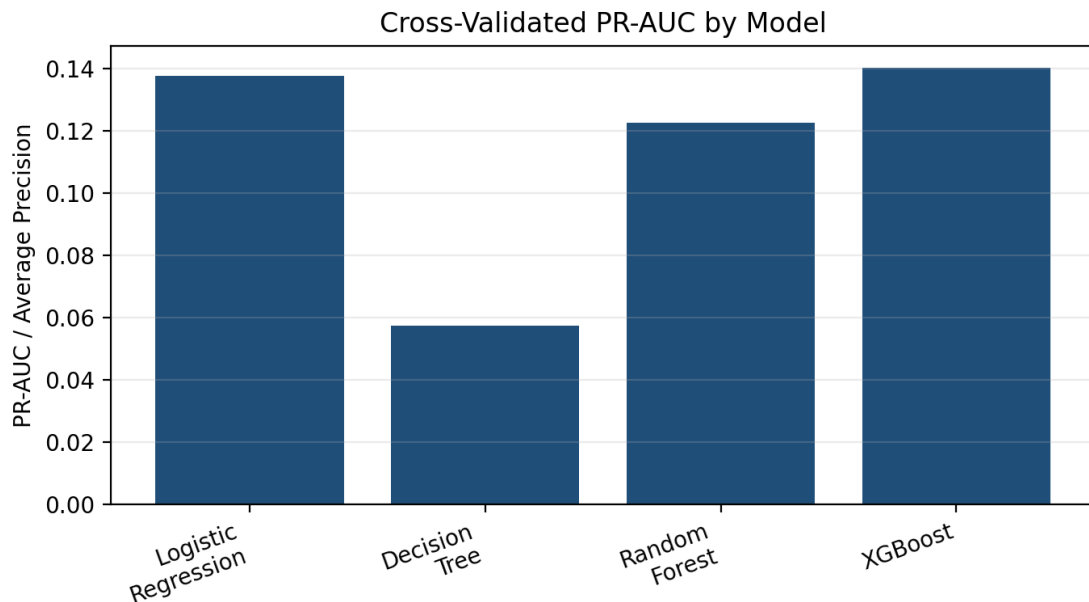


Figure 1. Cross-validated PR-AUC comparison across candidate models.

XGBoost achieved the highest cross-validated PR-AUC. However, the independent month 6 validation set produced a different model ranking. Logistic Regression achieved the highest validation PR-AUC of 0.164799 compared with 0.154837 for XGBoost and was selected as the provisional champion. This demonstrates why temporal validation was prioritized over cross-validation alone.

Stage	Metric	Result
Validation	PR-AUC	0.164799
Validation	Recall at Top 10%	62.90%
Validation	Precision at Top 10%	8.43%
Validation	Estimated Net Benefit	5,562,050
Final Base Test	PR-AUC	0.198539
Final Base Test	ROC-AUC	0.890668
Final Base Test	Fixed-threshold Recall	62.75%
Final Base Test	Fixed-threshold Precision	11.41%
Final Base Test	Recall at Top 10%	67.02%
Final Base Test	Precision at Top 10%	9.88%
Final Base Test	Estimated Net Benefit	6,419,500

Table 6. Champion Model Validation and Final Test Results

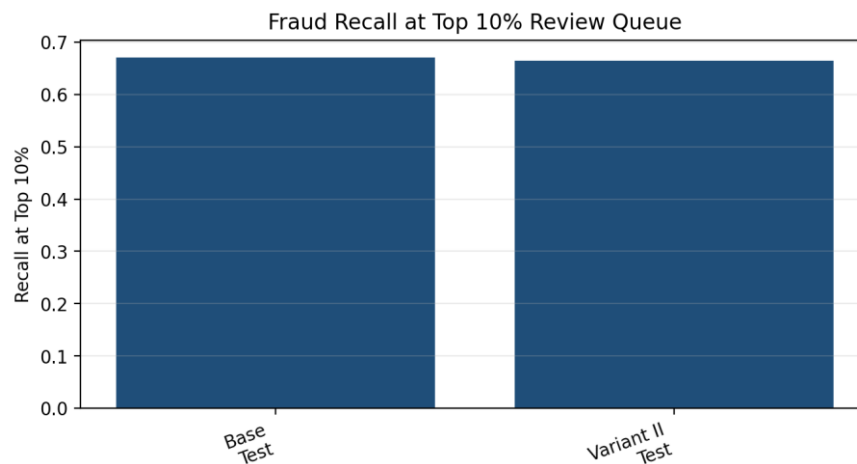


Figure 2. Fraud captured within the exact top 10% review queue on Base and Variant II.

After refitting on all Base training records, the Logistic Regression champion used a validation-derived threshold of 0.658130. On the unseen Base month 7 test set, it captured 67.02% of fraud within the top 10% highest-risk applications. The Base PR-AUC 95% bootstrap confidence interval was 0.182275 to 0.220415.

Metric	Base Month 7	Variant II Month 7	Change
PR-AUC	0.198539	0.201282	+0.002743
ROC-AUC	0.890668	0.889638	-0.001030
Recall at Top 10%	0.670168	0.664566	-0.005602
Precision at Top 10%	0.098813	0.097987	-0.000826
Estimated Net Benefit	6,419,500	6,337,500	-82,000

Table 7. Robustness Test: Base vs. Variant II

The Variant II stress test showed stable ranking performance, with PR-AUC of 0.201282 and top 10% recall of 66.46%. However, the unchanged Base threshold produced higher fixed-threshold recall but lower precision, a higher false-positive rate, and a worse Brier score. Therefore, the model is reliable as a risk-ranking tool but requires calibration and threshold review before operational use.

5. CRITICAL THINKING, ETHICAL AI, AND BIAS AUDITING

Step 5 evaluated whether the Logistic Regression champion is explainable, fair, stable, and governable for use as a fraud-review prioritization tool. The model was audited using coefficients, SHAP, local explanations, Partial Dependence Plots, Individual Conditional Expectation curves, leakage sensitivity analysis, calibration analysis, fairness metrics, threshold sensitivity, and a reweighting mitigation experiment.

5.1 Explainability Findings

The fitted coefficients confirmed several domain-relevant patterns. The strongest positive associations with predicted fraud included multiple emails on one device, unknown previous-address history, housing status BA, a negative intended-balance flag, credit-risk score, foreign request, free-email use, income, and customer age. The strongest negative associations included having other cards, a valid home phone, keep-alive sessions, log proposed credit limit, and 24-hour velocity.

Feature	Coefficient	Direction	
multiple_emails_per_device_flag	+1.628	Increases fraud risk	predicted
prev_address_months_count_unknown_flag	+1.503	Increases fraud risk	predicted
has_other_cards	-1.180	Decreases fraud risk	predicted
phone_home_valid	-0.992	Decreases fraud risk	predicted
housing_status_BA	+0.958	Increases fraud risk	predicted
intended_balcon_negative_flag	+0.823	Increases fraud risk	predicted
keep_alive_session	-0.677	Decreases fraud risk	predicted
log_proposed_credit_limit	-0.644	Decreases fraud risk	predicted
credit_risk_score	+0.625	Increases fraud risk	predicted
velocity_24h	-0.613	Decreases fraud risk	predicted

Table 8. Top Logistic Regression Coefficients

The SHAP analysis broadly supported the coefficient findings. Four-week velocity, unknown previous-address history, 24-hour velocity, phone validity, proposed credit limit, velocity ratios, card ownership, session persistence, credit-risk score, negative intended balance, free-email use, housing status, income, and age were among the strongest model drivers. These findings are predictive associations and should not be interpreted as causal statements.

Outcome Type	Count at Final Threshold
True Negative	88,456
False Positive	6,959

Outcome Type	Count at Final Threshold
True Positive	896
False Negative	532

Table 9. Base Test Confusion Matrix Outcomes at the Champion Threshold

The local case explanations included true positives, false positives, false negatives, and true negatives. This is important because a responsible fraud model must explain not only correct detections but also customer-friction cases and missed-fraud cases. The presence of false positives and false negatives reinforces the need for human analyst review.

5.2 Leakage, Calibration, and Threshold Stability

A potential preprocessing leakage concern was identified for `credit_risk_band` because it was created using dataset-wide quantiles before the temporal split. A targeted sensitivity test refitted the champion without this feature. Removing it had negligible effect on Base PR-AUC and slightly improved exact top 10% recall on Base, supporting removal or reconstruction using training-only cut points in a final production retraining.

Model Version	Dataset	PR-AUC	Recall at Top 10%	Precision at Top 10%	Brier Score
Original champion	Base month 7	0.198539	0.670168	0.098813	0.102045
Without <code>credit_risk_band</code>	Base month 7	0.198440	0.672969	0.099226	0.102231
Original champion	Variant II month 7	0.201282	0.664566	0.097987	0.144208
Without <code>credit_risk_band</code>	Variant II month 7	0.200398	0.656863	0.096851	0.141752

Table 10. Leakage Sensitivity Results

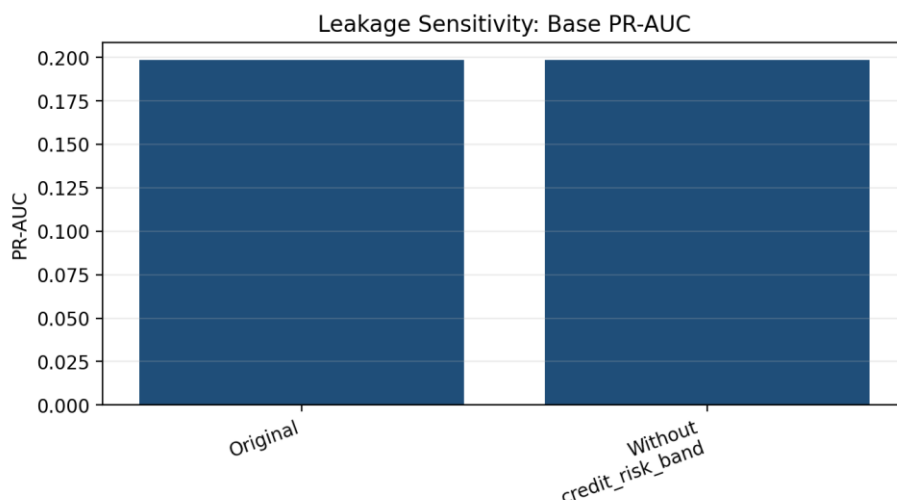


Figure 3. Removing `credit_risk_band` had negligible impact on Base PR-AUC.

The calibration analysis showed that the champion ranks applications effectively, but its raw probabilities are not calibrated. Mean predicted probability was 22.99% on Base and 28.60% on Variant II, while observed fraud prevalence was only 1.47% in both. Expected calibration error increased from 0.215 on Base to 0.271 on Variant II. This is consistent with the use of balanced class weights: the scores are useful for ordering risk but should not be interpreted as literal fraud probabilities without post-hoc calibration.

Dataset	Brier Score	Expected Calibration Error	Mean Predicted Probability	Observed Fraud Rate
Base month 7	0.102045	0.215144	22.99%	1.4746%
Variant II-month 7	0.144208	0.271248	28.60%	1.4746%

Table 11. Probability Calibration Summary

5.3 Fairness Audit

The primary fairness audit used age and income as available group attributes, with employment and housing included as supplemental socioeconomic proxy groups. Race, gender, ethnicity, disability, and other protected characteristics were unavailable and were not inferred. The fairness metrics are diagnostic measures rather than automatic legal pass-or-fail rules.

Fairness Area	Key Result
Age selection rate	Selection increased from 2.95% for applicants under 20 to 26.59% for ages 50-59; smaller older groups had even higher but less stable rates.
Age true-positive rate	TPR rose from 36.69% for applicants under 20 to 83.21% for ages 50-59.
Age fairness gaps	Demographic-parity difference was 0.334, disparate-impact ratio was 0.081, and equalized-odds difference was 0.513.
Income selection rate	Higher-income band had 10.19% selection rate versus 3.17% in the lower-income band.
Income true-positive rate	Higher-income band TPR was 65.54% versus 48.32% in the lower-income band.
Interpretation	Observed base-rate differences explain part, but not all, of the outcome gaps.

Table 12. Fairness Audit Highlights

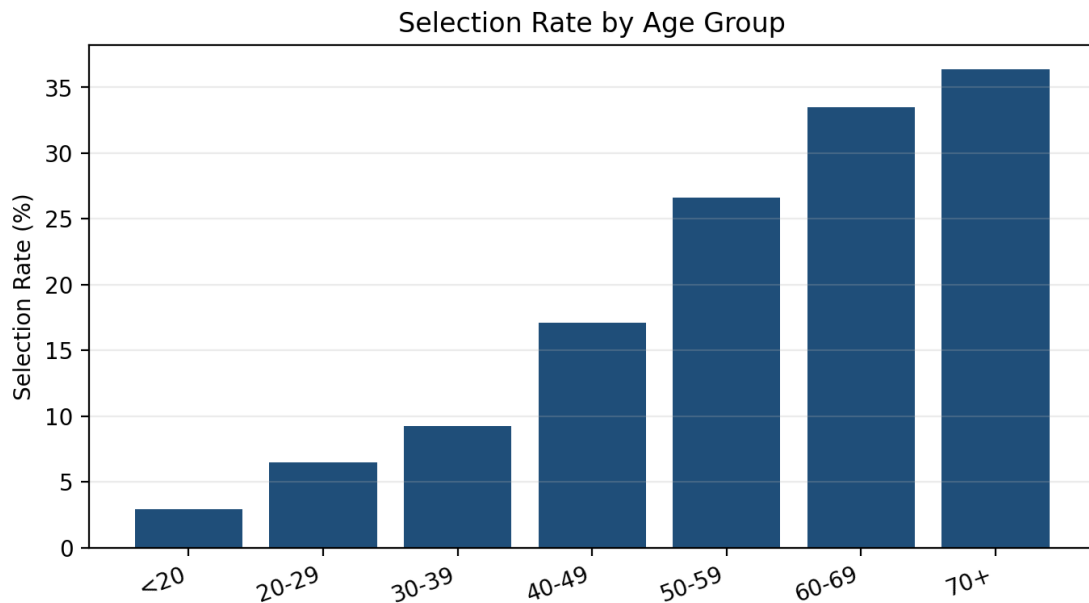


Figure 4. Selection rate differed materially across age groups.

The fairness audit identified material group differences across age and socioeconomic proxies. Employment and housing groups showed even larger gaps, although some categories had small fraud counts and require cautious interpretation. The results do not prove unlawful discrimination; they demonstrate that the model’s impacts are uneven and require governance review before deployment.

5.4 Review-Rate Sensitivity and Mitigation

The review-rate analysis evaluated 5%, 10%, 15%, and 20% policies using thresholds derived only from validation data. The 10% policy produced the highest illustrative net benefit and captured 67.02% of fraud in the exact top 10% review queue. A 15% policy improved recall and reduced the age equalized-odds gap but lowered net benefit due to higher review and false-positive cost.

Target Review Rate	Threshold	Recall at Top-K	Precision at Top-K	Estimated Net Benefit	Age Equalized-Odds Gap
5%	0.793884	51.40%	15.16%	5.828M	0.650
10%	0.658130	67.02%	9.88%	6.420M	0.513
15%	0.548891	75.98%	7.47%	6.037M	0.389
20%	0.462051	81.65%	6.02%	5.172M	0.486

Table 13. Threshold and Review-Rate Sensitivity

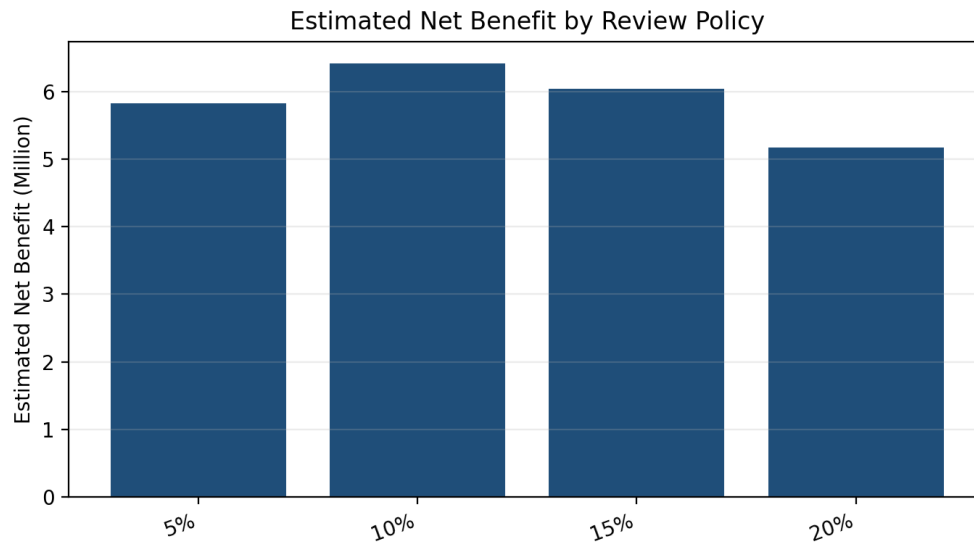


Figure 5. The 10% review policy produced the highest illustrative net benefit.

The tested age-income reweighting strategy was not adopted. It reduced validation PR-AUC from 0.165378 to 0.147608, recall from 64.69% to 61.52%, and top 10% lift from 6.47 to 6.15. Although age demographic-parity difference improved, age equal-opportunity difference worsened, and income fairness measures generally deteriorated. This confirms that mitigation must be judged across multiple technical and fairness metrics, not a single parity measure.

BUSINESS IMPACT AND OPERATING POLICY

The project translated model results into business terms using illustrative assumptions for estimated fraud loss, manual review cost, and false-positive cost. These assumptions are for capstone demonstration and must be replaced with institution-specific values before a real business case is approved.

Business KPI	Base Month 7 Result
Applications reviewed at 10% policy	9,685
Fraud captured in top 10% queue	957
Recall at top 10%	67.02%
Precision at top 10%	9.88%
Lift at top 10%	6.70x
Estimated fraud loss avoided	9,570,000
Estimated manual review cost	968,500
Estimated false-positive cost	2,182,000
Estimated net benefit	6,419,500

Table 14. Business Impact on 10% Review Capacity

The model materially improves fraud-review prioritization by concentrating approximately two-thirds of fraud into the highest-risk 10% of applications. This does not mean that the other 90% can be ignored; rather, it shows that the model can help analysts focus on limited investigative capacity more effectively.

LIMITATIONS

- Synthetic-data limitation: The BAF suite is useful for controlled experimentation but does not fully represent the customer base, controls, fraud typologies, products, and operating environment of a specific financial institution.
- Severe class imbalance: Fraud prevalence is low and changed across months, affecting precision, calibration, and review workload.
- Threshold transfer risk: The same threshold produced different workloads and false-positive rates on Base and Variant II.
- Calibration limitation: Balanced class weights improved ranking but produced probabilities that were much higher than observed fraud prevalence.
- Fairness-scope limitation: Race, gender, ethnicity, disability, and other protected characteristics were unavailable and were not inferred.
- Proxy-variable risk: Age, income, employment, and housing may influence review burden and require oversight of governance.
- Potential leakage concern: `credit_risk_band` was initially based on dataset-wide quantiles and should be removed or fit using training-only boundaries.
- Business-assumption limitation: Net-benefit values are illustrative and should be recalculated using validated institutional cost assumptions.

CHALLENGES ENCOUNTERED AND ADJUSTMENTS MADE

Challenge	Observed effect	Adjustment	Outcome
Large data volume and Colab memory pressure	Two 1-million-row files, feature expansion to 53 columns, one-hot encoding and explainability increased memory demand.	Used a 150,000-row stratified development sample for tuning; limited SHAP/PDP samples; retained full data for final champion refit and test.	Enabled completion without changing final temporal test design.
Logistic Regression non-convergence in the initial attempt	The earlier solver reached <code>max_iter</code> and produced repeated convergence warnings after a long run.	Changed to <code>newton-cg</code> , focused the C grid on five values, retained robust scaling, and monitored <code>n_iter_</code> .	Converged with <code>C=0.02</code> in 13/1000 iterations; tuning time reduced to 261 seconds with no warning.
Excessive hyperparameter-search time	Random Forest tuning took about 5,543 seconds;	Used <code>GridSearchCV</code> only for small spaces and <code>RandomizedSearchCV</code> with	Preserved comparative evidence while controlling

	exhaustive search would be impractical.	eight candidates and three folds for large spaces.	runtime.
Timeout and session interruption risk	Long Colab runs can disconnect or lose in-memory objects.	Saved CV result CSVs, models, configs and reports after each major stage; used fixed seeds and resumable notebook sections.	Reduced the cost of reruns and supported reproducibility.
SHAP computational warning	SHAP automatically reduced the 500-row background to 100 samples.	Accepted the controlled subsampling and limited evaluation to 1,500 records.	Produced stable global and local explanations without full-dataset cost.
Potential preprocessing leakage	credit_risk_band used dataset-wide qcut boundaries before the temporal split.	Performed an explicit sensitivity refit without the feature.	Base PR-AUC changed by only - 0.000099 and top 10% recall slightly improved, supporting feature removal.
Threshold and calibration drift	The same threshold produced higher selection and false-positive rates on Variant II.	Separated exact top-K ranking metrics from fixed-threshold metrics and added Brier/ECE analysis.	Confirmed robust ranking but poor probability calibration and threshold portability.
Fairness mitigation trade-off	Simple age-income reweighting improved some parity measures but hurt other fairness and predictive metrics.	Compared original and reweighted candidates on validation data before adoption.	Rejected the mitigation rather than optimizing one fairness measure in isolation.

CONCLUSION

This capstone demonstrated that machine learning can effectively support the prioritization of potentially fraudulent bank account applications. Logistic Regression was selected as the provisional champion because it performed best on the independent validation set while remaining transparent, explainable, and easier to govern than more complex models.

On the Base test set, the model achieved a PR-AUC of 0.1985, ROC-AUC of 0.8907, and captured 67.02% of fraud within the top 10% highest-risk applications. These results show that

the model can concentrate a large share of fraud cases into a smaller review queue, improving the efficiency of manual investigation.

The model also showed stable ranking performance on Variant II, with a PR-AUC of 0.2013 and top 10% fraud recall of 66.46%. However, differences in threshold behavior and calibration indicate that the model is more appropriate as a fraud-risk ranking tool than as an automatic approval or rejection mechanism.

RECOMMENDATIONS

The model should be used as a human-supervised decision-support tool that helps fraud analysts prioritize high-risk applications. Final decisions should remain with authorized reviewers, and model outputs should be supported by understandable reason codes or explainability results.

Before operational use, the model should be validated using actual banking data and recalibrated based on local fraud prevalence, review capacity, and cost assumptions. The `credit_risk_band` leakage concern should also be corrected, and fairness, calibration, and model drift should be monitored regularly.

XGBoost may be explored further as an alternative because it achieved the highest cross-validated PR-AUC. However, any performance improvement should be balanced against explainability, governance requirements, computational cost, and operational complexity.

Priority	Recommendation	Rationale
1	Validate on institution-specific historical applications	Confirms whether dataset patterns transfer to the real operating environment.
2	Remove or training-fit <code>credit_risk_band</code>	Addresses the identified leakage and consistency concern.
3	Calibrate probabilities and review thresholds periodically	Improves score interpretability and workload stability.
4	Deploy with analyst review and reason codes only	Reduces the risk of automated unfair or incorrect decisions.
5	Monitor fairness, drift, calibration, and business KPIs monthly	Ensures ongoing responsible use as fraud behavior and populations change.
6	Evaluate calibrated XGBoost or ensemble alternatives	May improve performance if governance and explainability needs are satisfied.

Table 15. Prioritized Improvement Roadmap

REFERENCES

- Jesus, S. (2022). Bank Account Fraud Dataset Suite (NeurIPS 2022) [Data set]. Kaggle. <https://www.kaggle.com/datasets/sgpjesus/bank-account-fraud-dataset-neurips-2022>
- Jesus, S., Pombal, J., Alves, D., Cruz, A., Saleiro, P., Ribeiro, R. P., Gama, J., & Bizarro, P. (2022). Turning the Tables: Biased, Imbalanced, Dynamic Tabular Datasets for ML Evaluation. Advances in Neural Information Processing Systems, 35. https://proceedings.neurips.cc/paper_files/paper/2022/hash/d9696563856bd350e4e7ac5e5812f23c-Abstract-Datasets_and_Benchmarks.html
- Feedzai. (2022). Bank Account Fraud Dataset Suite. GitHub. <https://github.com/feedzai/bank-account-fraud>
- Scikit-learn documentation. <https://scikit-learn.org/stable/>
- XGBoost documentation. <https://xgboost.readthedocs.io/>
- SHAP documentation. <https://shap.readthedocs.io/>
- Fairlearn documentation. <https://fairlearn.org/>